Inertia and balanced forces

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Predict motion when the net force is zero.

SUCCESS CRITERIA - I can...

1. Add forces to decide whether they balance.
2. Predict constant velocity when net force is zero.
3. Explain why changing direction requires net force.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “net force” and “inertia” to explain your model or result. Sentence starters are optional.

net force: the vector sum of all forces on one object

inertia: the tendency to keep the same velocity without net force

balanced: adding to zero net force in this force context

01 Warm-up: bring back an earlier idea

A goes 6 m east and 4 m north. B then goes 3 m west and 8 m south.
Draw the route and find the endpoint relative to the start.

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: $\Sigma F = 0 \Rightarrow a = 0 \Rightarrow v = \text{constant}$.
ΣF means net force. Balanced forces allow rest or steady motion in a straight line.

Worked reasoning: A book has 6 N upward support and 6 N downward weight. Net force is zero. Its velocity stays constant; it could be resting or moving steadily.

Your turn: Draw the two force arrows. Then sketch two possible velocity-time graphs consistent with this balance.

Sentence starter: The ... in my model represents ... . Its label shows ... .



Book: compare support and weight.

Marzano check - circle: ① ② ③

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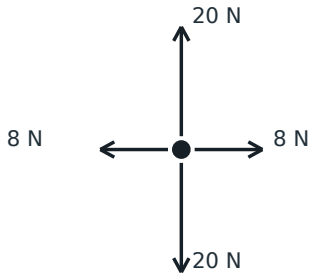
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Sentence starters are optional. You may plan in any language.

03 Balanced while moving

The box is sliding right at steady velocity. Add the horizontal forces shown. Does zero net force mean the box must stop?

Sentence starter: I notice … . This supports … because … .



The dot represents the object. Box moving right

Marzano check - circle: 1 2 3

04 After release

You push a puck, then your hand loses contact. Assume a level surface with no friction or air resistance. Complete the puck's force diagram after release. What happens to the hand force and to the puck's horizontal velocity?

Sentence starter: The … in my model represents … . Its label shows … .

Add and label force arrows.

Puck after release; it is moving right.

Marzano check - circle: 1 2 3

05 The bus stops

A bus brakes while traveling forward.
Explain why a passenger tends to keep moving forward relative to the bus.
Draw the passenger's velocity and a separate arrow for the force from a seat belt.

Sentence starter: My claim is … . The diagram/data show …, so … .

Passenger during braking

Forward is right.

Velocity:

Acceleration / force:

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

A cart accelerates uniformly from rest to 10 m/s east in 3 s.
Find acceleration, average velocity and displacement.
Sketch the velocity-time model; label the given values.

Word help: acceleration = rate of change of velocity

Sentence starter: I found … [unit]. This value means … in this situation.

Velocity (m/s)

Time (s)

Marzano check - circle: 1 2 3

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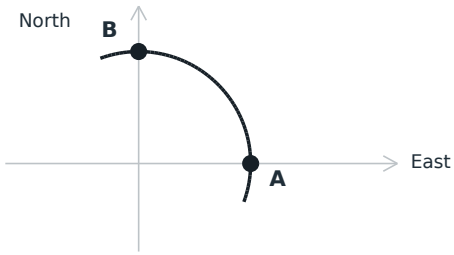
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Sentence starters are optional. You may plan in any language.

07 Exit check

Which motions can have zero net force: resting, straight at steady speed, turning at steady speed, or speeding up? For the turning case shown, draw velocity arrows at A and B and explain your choice.

Sentence starter: My claim is … . The diagram/data show …, so … .



Path sketch: runner moves from A toward B.

Marzano check - circle: ① ② ③

08 STRETCH: test the model

A puck slides right at steady speed.
Draw one possible balanced force diagram and explain why a forward force is not required in the frictionless model.

Sentence starter: My claim is … . The diagram/data show …, so … .



Add and label force arrows.

Puck moving right on a frictionless level surface.

Marzano check - circle: ① ② ③